Shirisha Gujja

San Jose, CA | shirishagujja456@gmail.com | +1 (669) 360-9706 | GitHub | LinkedIn | Portfolio

Education

San Jose State University — Jan 2026 – Present
Master of Science in Software Engineering — San Jose, CA

Vasavi College of Engineering — Dec 2020 – May 2024
Bachelor of Engineering in Computer Science — Hyderabad, India

Skills

- **Programming Languages:** Python, Java, TypeScript, JavaScript, SQL, C
- **Frameworks & Technologies:** FastAPI, Django, React.js, Next.js, Node.js, Express.js, SQLAlchemy, REST APIs, GraphQL
- **Databases, Cloud & DevOps:** PostgreSQL, MySQL, MongoDB, MS SQL Server, Redis, AWS (EC2, S3, IAM, SES), Docker
- **AI & Tools:** OpenAI GPT-4, Claude, GitHub Copilot, Pandas, NumPy, Scikit-learn, OpenCV

Experience

S&P Global — Sep 2024 – Dec 2025
Software Development Engineer — Hyderabad, India

- Designed and optimized **MCP-based tool integrations** for LLM-powered enterprise features, enabling structured GraphQL data retrieval and context-aware application workflows while improving **tool execution latency by 25%** and increasing model-to-backend interaction reliability.
- Built **selective GraphQL retrieval mechanisms** for LLM tool calls, reducing unnecessary data transfer by **35%** and decreasing **API response times by 20%** through targeted field selection and optimized query execution.
- Developed reusable React components and AI-powered UI widgets for displaying model-generated insights and backend data, reducing duplicate frontend development effort across multiple enterprise workflows.
- Trained and configured Copilot using historical support data, automating repetitive support tasks and **reducing manual workflow effort by 40%**, enabling support teams to resolve common requests more efficiently.
- Integrated LLMs, MCP, GraphQL, REST APIs, and React across enterprise workflows, contributing to a **25% improvement in end-to-end AI feature** response performance through optimized data retrieval and application-layer interactions.

S&P Global — Jan 2024 – Jul 2024
Software Development Engineer Intern — Hyderabad, India

- Led a proof-of-concept to re-architect a critical daily refresh job from **MS SQL Server** stored procedures to **Amazon Redshift** using **PostgreSQL**-based stored procedures, improving scalability and query performance.
- Engineered modular **Python ETL** workflows to orchestrate **Redshift** data-processing pipelines, incorporating automated data validation, logging, error handling, and retry mechanisms, with **AWS Lambda** supporting workflow automation.
- Reduced end-to-end refresh time from **9 hours to 1 hour 27 minutes**, achieving an approximately **84% reduction in execution time**, and presented the optimized architecture for adoption as the standard production refresh process.

Projects

Personal Finance Platform — GitHub

- Built a finance platform with Next.js, TypeScript, FastAPI, PostgreSQL, and Redis for accounts, transactions, budgets, savings goals, investments, net worth, and cash flow forecasting with user-scoped authorization and authentication.
- Architected an event-driven transaction processing pipeline using a transactional outbox and a Redis-backed worker to categorize transactions, detect unusual spending, and deliver real-time alerts, improving processing reliability.
- Integrated Plaid for secure bank account linking and transaction synchronization, implementing encrypted token storage and per-user authorization for sensitive financial data.
- Implemented OpenAI-based merchant categorization and monthly spending insights with JSON schema validation, rate limit handling, and rule-based fallbacks to ensure robust insight generation and graceful degradation.

Fitness Tracker — Website | GitHub

- Built and deployed Fitness Tracker, a full stack fitness platform using Next.js, FastAPI, PostgreSQL, and Docker, enabling users to manage workouts, nutrition, body metrics, and fitness goals through a modular domain driven architecture.
- Developed six backend domains covering authentication, workouts, nutrition, progress tracking, profiles, and AI coaching with dedicated models, validation schemas, service layers, and REST API endpoints for scalable feature development.
- Implemented secure authentication using JWT access tokens, refresh token rotation, HTTP only cookies, password hashing, session revocation, and automatic token refresh to ensure reliable and protected user sessions.
- Integrated the OpenAI API to generate personalized workout and meal plans and provide context aware fitness coaching with structured response validation, rate limit handling, and graceful fallbacks for consistent user experience.